

KAHRAMANMARAS, Turkey

WMO: 172550

Lat: 37.600N

Lon: 36.933E

Elev: 572

StdP: 94.64

Time Zone: 3.00 (E03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	-0.8	-13.2	1.3	4.5	-10.2	1.7	6.5	9.3	6.7	8.4	6.6	1.6	80	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.6	39.1	20.7	37.8	20.3	36.1	20.5	25.5	32.6	24.4	31.4	23.5	30.8	2.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.8	29.7	22.5	18.4	28.9	21.2	17.0	28.3	82.6	32.7	77.6	31.2	73.4	30.7	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.7	8.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	5.3	7.6	12.0	16.3	20.4	25.9	29.4	29.7	26.1	19.7	12.8	7.8
	DBStd	8.84	2.84	3.16	3.12	3.20	2.95	2.29	1.84	1.76	2.58	3.10	3.28	2.63
	HDD10.0	331	146	80	16	1	0	0	0	0	0	0	14	74
	HDD18.3	1508	404	301	198	78	14	0	0	0	1	22	165	327
	CDD10.0	3180	1	12	77	189	322	476	603	612	484	299	99	6
	CDD18.3	1316	0	0	1	16	78	226	344	353	234	63	0	0
	CDH23.3	15545	0	1	5	189	769	2407	4510	4571	2549	539	6	0
	CDH26.7	7976	0	0	1	41	266	1149	2554	2547	1266	153	0	0
Wind	WSAvg	2.5	1.4	1.4	2.0	2.0	2.4	4.2	4.8	4.6	3.3	1.6	1.0	1.6
Precipitation	PrecAvg	694	117	109	88	71	45	8	2	2	15	45	81	116
	PrecMax	1081	312	208	249	168	141	29	8	6	94	132	239	288
	PrecMin	431	14	21	10	17	11	0	0	0	0	2	0	0
	PrecStd	145	73	53	52	37	28	7	2	2	19	31	56	71
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	20.6	23.7	30.3	34.7	38.3	41.2	41.0	39.0	32.2	24.1	18.1
		MCWB	7.9	11.0	13.7	15.7	18.0	20.0	21.2	20.4	18.2	17.9	12.9	10.5
	2%	DB	12.2	16.8	21.3	27.4	31.9	36.1	39.2	39.0	36.9	30.2	22.3	15.3
		MCWB	6.8	9.8	12.0	14.6	17.0	19.6	20.7	21.5	17.9	17.1	12.2	8.5
	5%	DB	10.9	14.9	20.0	25.2	29.5	34.2	37.8	37.7	34.9	28.2	20.8	13.8
		MCWB	6.4	8.8	11.4	14.3	16.6	19.1	20.1	21.3	18.2	16.1	12.0	8.1
	10%	DB	9.8	13.0	18.2	23.5	27.6	32.4	36.1	36.0	32.9	26.2	19.0	12.0
		MCWB	6.2	7.8	10.7	13.7	15.8	19.0	20.8	21.4	18.7	15.2	11.2	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	11.8	15.0	18.2	20.1	23.9	26.7	26.8	24.2	21.9	15.6	12.1
		MCDB	11.9	18.6	21.4	27.9	29.2	31.1	34.2	34.1	29.4	25.9	20.8	15.2
	2%	WB	8.3	10.5	13.4	16.5	18.9	22.8	25.7	25.7	22.9	19.7	13.8	10.4
		MCDB	10.7	15.6	19.2	24.0	27.2	31.5	32.9	32.7	29.0	26.2	19.4	13.6
	5%	WB	7.6	9.5	12.4	15.3	17.9	21.7	24.6	24.7	22.0	18.0	13.0	9.6
		MCDB	9.8	13.8	17.7	22.2	26.5	30.2	31.7	31.7	28.4	24.7	18.1	11.9
	10%	WB	6.7	8.6	11.5	14.5	17.1	20.8	23.5	23.7	21.1	16.9	12.1	8.8
		MCDB	8.8	12.1	16.5	21.3	25.5	28.9	31.0	31.0	27.9	23.7	17.3	11.0
Mean Daily Temperature Range	5% DB	MDBR	6.8	8.5	10.1	11.6	12.3	12.7	14.0	13.6	13.9	12.4	11.2	7.7
		MCDBR	9.8	11.6	13.0	15.0	15.9	15.5	17.0	16.2	16.9	14.9	14.0	11.6
		MCWBR	5.4	5.7	5.7	5.7	5.3	4.9	5.7	4.9	5.8	5.7	6.4	6.4
	5% WB	MCDBR	7.0	9.8	11.1	12.9	14.2	13.4	14.0	13.7	13.7	12.7	12.0	8.7
MCWBR		4.4	5.0	5.2	5.3	5.2	4.8	4.7	5.1	5.2	5.4	6.0	5.5	
Clear-Sky Solar Irradiance	taub	0.341	0.367	0.406	0.452	0.430	0.383	0.377	0.392	0.363	0.392	0.353	0.335	
	taud	2.295	2.225	2.151	2.055	2.181	2.343	2.368	2.308	2.387	2.270	2.340	2.358	
	Ebn at Noon	826	850	850	830	855	895	897	876	887	813	808	809	
	Edh at Noon	97	118	140	164	147	125	121	126	110	112	93	86	
All-Sky Solar Radiation	RadAvg	2.13	2.97	4.24	5.36	6.52	7.78	7.87	6.99	5.73	3.95	2.76	2.02	
	RadStd	0.35	0.49	0.36	0.51	0.38	0.36	0.25	0.24	0.25	0.29	0.29	0.36	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page